

SCS 1302 - Discrete Mathematics

Take Home Assignment 02

Release Date	03 August 2026	Duration	12 Days
Deadline	15 August 2026, 11.59 PM	Raw Marks	200 (Scaled to 100)

Instructions

- Attempt all four questions. The total raw mark is 200 and will be scaled to 100.
- Handwritten or typed answers are allowed; however, every answer must show complete reasoning, logical steps, and justification.
- Submit one PDF file only. Write your index number on every page and rename the file as Index_A2 (example: 240000_A2).
- Where a statement or property is false, provide a suitable counterexample.
- Clearly state every theorem, definition, identity, or result used in your solution.
- Late submissions will not be marked. Submit on or before **15 August 2026 at 11.59 PM**.

Question 01:

(50 marks)

(a) Let

$$U = \{1, 2, 3, \dots, 20\}, \quad A = \{x \in U : 2 \mid x\}, \quad B = \{x \in U : 3 \mid x\},$$

$$C = \{x \in U : x \text{ is prime}\}.$$

Determine, in roster form,

$$(A \cup B)^c, \quad (A \cap B) \cup C, \quad (A - C) \triangle (B - C),$$

and

$$((A \cup C) \cap (B \cup C)) - C.$$

Hence verify, for these particular sets, that

$$(A \cup C) \cap (B \cup C) = (A \cap B) \cup C.$$

(10 marks)

(b) Let $X = \{a, b, c, d, e\}$. Without listing the entire power set, determine the number of subsets of X that

- (i) contain a ;
- (ii) contain both a and b ;
- (iii) contain exactly three elements;
- (iv) contain a but do not contain e .

Explain the counting argument used in each case.

(10 marks)

(c) Prove, using the element-membership method, that for arbitrary sets A , B , and C ,

$$A - (B \cup C) = (A - B) \cap (A - C).$$

Your proof must establish both set inclusions. **(10 marks)**

- (d) Let $U = \{1, 2, \dots, 12\}$, with its elements in increasing order. The subsets A , B , and C are represented by

$$A : 101101001011, \quad B : 011011100101, \quad C : 110010011010.$$

Obtain the bit-string representation and roster form of

$$(A \cap B^c) \cup C \quad \text{and} \quad (A \Delta B) \cap C^c.$$

Show every bitwise operation used. **(10 marks)**

- (e) Use set identities to simplify

$$[(A \cup B) \cap (A \cup B^c)] \cup [A^c \cap (B \cup B^c)].$$

Name each identity used. Then verify the simplified result using an element-membership argument. **(10 marks)**

Question 02: **(50 marks)**

- (a) Let $A = \{-2, -1, 0, 1, 2\}$ and define $f : A \rightarrow \mathbb{Z}$ by

$$f(x) = x^2 - x.$$

Find the range of f and determine whether f is one-to-one. Next, let $B = f(A)$ and regard f as a function $f : A \rightarrow B$. Determine whether this new function is onto and whether it is bijective. Justify every conclusion. **(10 marks)**

- (b) Define

$$f : \mathbb{R} \setminus \{-1\} \longrightarrow \mathbb{R} \setminus \{2\}, \quad f(x) = \frac{2x-3}{x+1}.$$

Prove that f is one-to-one and onto. Hence find f^{-1} and verify one of the identities

$$f^{-1}(f(x)) = x \quad \text{or} \quad f(f^{-1}(y)) = y.$$

(10 marks)

- (c) Let $f, g : \mathbb{Z} \rightarrow \mathbb{Z}$ be given by

$$f(x) = 2x - 1, \quad g(x) = x^2 + 2.$$

Find $(f \circ g)(x)$, $(g \circ f)(x)$, and $(f \circ f)(x)$. Determine all integers x , if any, for which

$$(f \circ g)(x) = (g \circ f)(x).$$

(10 marks)

- (d) Let $f : A \rightarrow B$, and let $S, T \subseteq A$.

(i) Prove that $f(S \cup T) = f(S) \cup f(T)$.

(ii) Prove that $f(S \cap T) \subseteq f(S) \cap f(T)$.

(iii) Give a finite counterexample showing that equality in part (ii) need not hold when f is not one-to-one.

(10 marks)

(e) Evaluate

$$\left\lfloor -\frac{17}{4} \right\rfloor, \quad \left\lceil -\frac{17}{4} \right\rceil, \quad \lfloor \sqrt{20} \rfloor, \quad \lceil \sqrt{20} \rceil.$$

Then prove that, for every real number x and every integer n ,

$$\lfloor x + n \rfloor = \lfloor x \rfloor + n.$$

(10 marks)

Question 03:

(50 marks)

(a) Let $A = \{1, 2, 3, 4, 6, 12\}$ and define a relation R on A by

$$a R b \iff a \mid b.$$

List all ordered pairs in R . Determine whether R is reflexive, symmetric, anti-symmetric, and transitive. Give a proof or a counterexample for each property. (10 marks)(b) Define a relation S on $\mathbb{Z}$ by

$$a S b \iff a - b \text{ is divisible by } 4.$$

Prove that S is reflexive, symmetric, and transitive. Is it anti-symmetric? Justify your answer with either a proof or a counterexample. (10 marks)

(c) Let

$$R = \{(1, a), (1, b), (2, b), (3, c), (4, a)\}$$

be a relation from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c\}$, and let

$$S = \{(a, x), (a, z), (b, y), (c, x), (c, z)\}$$

be a relation from B to $C = \{x, y, z\}$. Find $S \circ R$. Then determine whether R and $S \circ R$ are functions from their respective domains to codomains. Explain. (10 marks)

(d) Let

$$R = \{(1, 2), (2, 3), (3, 1), (3, 4), (4, 4)\}$$

be a relation on $A = \{1, 2, 3, 4\}$. Compute R^2 and R^3 . Use your calculations to determine whether R is transitive. Identify one ordered pair that demonstrates your conclusion. (10 marks)(e) Let A be a set with four elements.

- (i) How many different relations can be defined on A ?
- (ii) How many reflexive relations can be defined on A ?
- (iii) How many symmetric relations can be defined on A ?

Give a complete counting argument. (10 marks)

Question 04:

(50 marks)

(a) Prove directly that if $a \mid b$ and $a \mid c$, then

$$a \mid (5b - 3c),$$

where $a, b, c \in \mathbb{Z}$ and $a \neq 0$. Your proof must begin from the definition of divisibility. (10 marks)(b) Prove by contradiction that $\sqrt{3}$ is irrational. You may use the fact that if $3 \mid n^2$, then $3 \mid n$. Clearly identify the contradiction obtained. (10 marks)

- (c) Use the Euclidean algorithm to find

$$\gcd(414, 662).$$

Then use back-substitution to express the greatest common divisor in the form

$$414s + 662t$$

for integers s and t .

(10 marks)

- (d) Solve the linear congruence

$$18x \equiv 30 \pmod{42}.$$

State how many incongruent solutions exist modulo 42, find all of them, and verify each solution.

(10 marks)

- (e) Let p be a prime number. Prove by induction on $n \geq 1$ that

$$p \mid a^n \implies p \mid a$$

for every integer a . Then use this result in a proof by contradiction to show that $\sqrt{p}$ is irrational. Clearly identify the contradiction obtained under the assumption that $\sqrt{p}$ is rational in lowest terms.

(10 marks)

End of Assignment